Giacomo Pasin

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Education

ETH Zürich Sep 2026 – Present
MSc in Computer Science *Zürich, Switzerland*

- Major: Machine Intelligence; Minor: Data Management Systems

Eindhoven University of Technology Sep 2023 – Jul 2026
BSc in Computer Science and Engineering *Eindhoven, Netherlands*

- Graduated *Cum Laude*
- Thesis: *Explainable RF Fingerprinting*. Reproduced and evaluated two spectrogram-based ML pipelines and conducted a partial ablation study to investigate misconfiguration.

Experience

AI Engineering and Software Development Intern Jun 2026 – Aug 2026
Mea World *Pistoia, Italy*

- Improved source selection in the RAG pipeline behind Kiuwo, so answers draw on more relevant documents
- Developed an internal AI agent that autonomously resolves the tickets assigned to it and reviews its own output against the original spec

CTO Aug 2021 – Present
My Twin Communication *Treviso, Italy*

- Consolidated infrastructure across two merging communication agencies and built internal tools supporting workflows across departments.

Foundations of Data Analytics Development Team Nov 2024 – May 2026
TU Eindhoven *Eindhoven, Netherlands*

- Built Jupyter notebook test frames and exercises for the course "Foundations of Data Analytics":
- Wrote test cases for the 1000+ student submissions

Selected Projects

CheckInOS | *TypeScript, Next.js, tRPC* 02/2025

- Enabled guest check-in on two mobile platforms (iOS and Android) by connecting native QR scanning to web-managed attendee rosters.
- Unified paid ticketing, live arrival tracking and on-site badge printing in one event workflow by integrating Stripe payments and registration-versus-arrival analytics.

ML for IR Spectroscopy | *Python, PyTorch, OpenCV* 06/2025

- Built a training corpus of 60,000+ IR spectra in under two days by developing two Selenium scrapers for NIST and SDBS and normalizing GIF and JCAMP-DX data.
- Extended functional-group prediction from pure compounds to binary mixtures by reproducing a published CNN in PyTorch and developing a fine-tuning pipeline using synthetic mixture spectra.
- Enabled spectrum uploads to return functional-group probabilities through a web interface by deploying the trained model behind a Flask API with Swagger documentation.

Research Terminal | *TypeScript, Next.js, Claude* 09/2026

- Enabled literature discovery through three entry points (title, DOI and natural-language ideas) by combining keyword and semantic search over OpenAlex.
- Made literature reviews resumable, with saved multi-paper citation graphs and per-paper AI summaries, by integrating reference/citation expansion, Claude summarization and account-based autosave.

Skills

ML & Data: Python, PyTorch, NumPy, scikit-learn, Pandas, OpenCV, Matplotlib, Jupyter Notebooks
Technologies: TypeScript, Git, Docker, Linux, Redis, MongoDB
Languages: English (Native), Italian (Native), Spanish (B2)